Day - 11

Quantum Entanglement

Today, we combine the Hadamard (H) gate and the Controlled-NOT (CNOT) gate to create a phenomenon that has no classical equivalent: **Quantum Entanglement**. This is arguably the most powerful resource in quantum computing.

1. What is Quantum Entanglement?

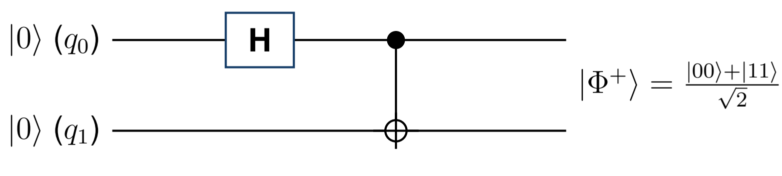
Quantum Entanglement is a phenomenon where two or more qubits become linked in such a way that they share a *single, combined quantum state*. You can no longer describe the state of one qubit independently of the other.

Classical Correlation vs. Quantum Entanglement

Classical Correlation: Like two synchronized clocks. They show the same time because of a shared history, but each clock has its own separate internal mechanism.
Quantum Entanglement: The qubits behave as one single quantum system observed from two different locations. Measuring one qubit instantly determines the correlated outcome of the pair, because they share the same mathematical description.

2. Building Entanglement: The Circuit

Entanglement isn't automatic; it must be built. The standard way to create the simplest entangled state (a Bell State) requires just two gates: an H gate to create superposition, and a CNOT gate to link the qubits.



Step-by-Step Breakdown:

1. **Initial State:** Both qubits start at $|0\rangle$. The combined state is $|00\rangle$.
2. **Apply Hadamard to q_0:** q_0 enters a 50/50 superposition. The combined state becomes $\frac{|00\rangle+|10\rangle}{\sqrt{2}}$. Notice q_1 is still $|0\rangle$.

3. **Apply CNOT (q_0 controls q_1):** The CNOT flips q_1 *only if* q_0 is 1. The $|00\rangle$ part stays $|00\rangle$. The $|10\rangle$ part becomes $|11\rangle$.
4. **Final Entangled State:** We have created the Bell State $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$.

3. Understanding the Output Statistics

If you measure the $|\Phi^+\rangle$ Bell state on an ideal quantum simulator 1000 times, you will see a unique statistical distribution:

Measured State	Theoretical Probability	Simulated Counts (Approx.)
$\|00\rangle$	50%	~ 500
$\|11\rangle$	50%	~ 500
$\|01\rangle$	0%	0
$\|10\rangle$	0%	0

You will **never** measure $|01\rangle$ or $|10\rangle$. The results are perfectly correlated. You cannot predict if the next shot will yield 00 or 11, but you know with absolute certainty that whatever value q_0 collapses to, q_1 will instantly collapse to the exact same value.

4. Qiskit Implementation

Creating this in Python is incredibly straightforward using the tools we learned on Day 10.

Python/Qiskit Code for Bell State

```
from qiskit import QuantumCircuit

# Create circuit with 2 qubits, 2 classical bits
qc = QuantumCircuit(2, 2)

# Apply H gate to qubit 0 (creates superposition)
qc.h(0)

# Apply CNOT gate (entangles the qubits)
qc.cx(0, 1)

# Measure both
qc.measure([0,1], [0,1])
```

5. Addressing Common Misconceptions

- **Myth:** *"Entanglement allows faster-than-light communication."*
  Correction: No. Entanglement creates correlations, but because the measurement outcome is random, it cannot be used to transmit usable information instantly.
- **Myth:** *"A CNOT gate always creates entanglement."*
  Correction: No. A CNOT gate only creates entanglement if the control qubit is

in a superposition (like after an H gate). If the control qubit is definitively $|0\rangle$ or $|1\rangle$, no entanglement occurs.